

THE LIFE CYCLES OF STARS

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STARS FORM WHEN clouds of interstellar gas collapse under the influence of gravity (see pp.238–239). During their lifetimes, stars pass through a series of stages, with the sequence and timing depending crucially on the mass of the star. As a star passes through these stages, different elements are created, again depending on the star's mass. When stars have completed their development, they shed their material back into the interstellar medium, enriching the matter from which future generations of stars will form.

LIFE STAGES
The environs of the nebula NGC 3603 display most stellar life stages, from "pregnant" dark nebulae and pillars of hydrogen, to a cluster of young stars, to a red star nearing its end.



DENSE CLOUDS START TO COLLAPSE

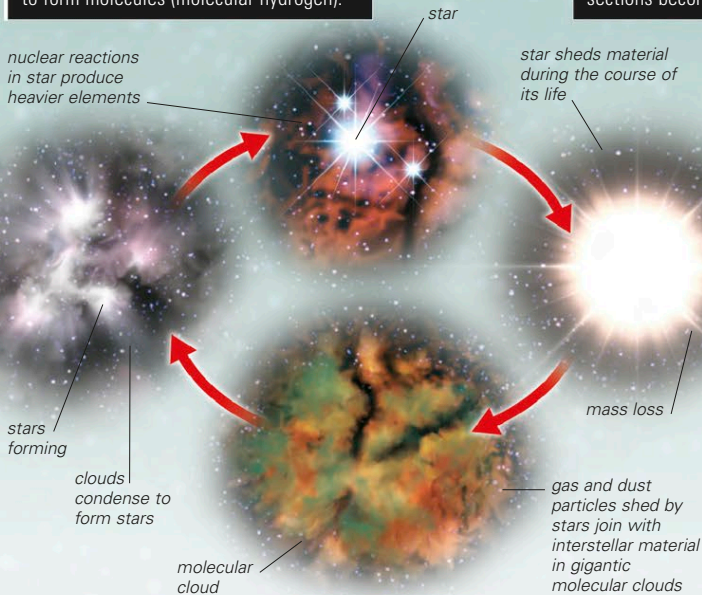
Stars form from cold interstellar clouds. The colder the cloud, the less resistant it is to gravitational collapse. Clouds are formed mostly of hydrogen. At low temperatures, hydrogen atoms combine to form molecules (molecular hydrogen).

PROTOSTARS BEGIN TO FORM

If the cloud is over a certain mass and it experiences a gravitational tug, it will begin to collapse. As it does, it will fragment into smaller parts of differing size and mass. These fragmented cloud sections become protostars.

PRESSURE AND TEMPERATURE RISE

The protostar continues to collapse, and the central temperature and pressure build up. The temperature and pressure levels will depend on the initial mass of the fragment—the higher the mass, the higher the temperature and pressure.



STAR-MAKING RECIPE

The basic ingredients of stars are found in cold clouds made mostly of hydrogen molecules. The early stages of star formation are initiated by gravity, which can be exerted by the tug of a passing object, a supernova shock wave, or the compression of one of the Milky Way's density waves. If the cloud has sufficient mass, it will collapse into a protostar, which contracts until nuclear reactions start in its core. At this point, a star is born. During its lifetime, a star will convert hydrogen to helium and a series of heavier materials, depending on its mass. These materials are gradually lost to the interstellar medium

ONGOING CYCLE
Stars form from material shed by previous generations of stars, and the death of massive stars can trigger the birth of others.

until the star has used up most of its fuel and begins to collapse. For a high-mass star, this will result in a supernova that scatters much of the remaining material into space.

BROWN DWARF

In protostars less than 0.08 solar masses, the pressure and temperature at the core do not get high enough for nuclear reactions to begin. These protostars become brown dwarfs.